

PRODUCT REQUIREMENTS DOCUMENT

Agentic AI Data Observability System

Clinical Trials — Real-Time Kafka Streaming

Version: 4.0 Enhanced

Domain: Life Sciences / Clinical Trials

Status: Authoritative Implementation Guide

Date: May 2025

Tech Stack: React 18 + TypeScript · Vite 6 · Tailwind CSS 3 · Lucide React · Recharts

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1. Project Overview & Architecture

1.1 Purpose

This document is the single authoritative specification for building the Agentic AI Data Observability System for Clinical Trials. A developer reading this document must be able to build every page without making assumptions. Every UI component, API endpoint, request body, response schema, and trigger condition is defined here.

1.2 System Summary

The platform provides real-time, AI-powered observability for clinical trial data pipelines. It accepts three input modes, runs pre-ingest validation, streams data through Apache Kafka, runs five specialized AI agents, and gates all results behind a Human-in-the-Loop review workflow.

1.3 High-Level Architecture

Layer	Components
Data Input	User Upload (CSV/JSON) Synthetic Generator External API Connector
Validation Gate	Pre-ingest preflight: hard blocks + soft warnings + cross-field rules
Streaming	Hospital API Simulator → Kafka Producer → Kafka Topic → Kafka Consumer
Observability Engine	Rolling Metrics: Freshness · Volume · Schema · Distribution · Lineage
PII/PHI Layer	Microsoft Presidio masks all PHI before agents see data
AI Agents (5)	Data Quality · Log Analysis · RCA · Recommendation · Compliance
HITL Gate	Clinical Data Manager approves/rejects — dashboard gated until approved
Frontend	React 18 + TypeScript + Vite 6 + Tailwind CSS 3 + Recharts
Storage	Local file system only — output/runs/{run_id}/ — no cloud, no Snowflake

1.4 Application Flow

- User authenticates (Login page)
- User navigates to Pipeline → selects input mode → uploads/generates/connects data
- Preflight Validation Gate runs → hard blocks must be resolved → user confirms
- User triggers Run Pipeline → events stream to Kafka in real time
- Streaming Pipeline page shows live Kafka metrics, lag, throughput, AI agent status
- AI agents analyze rolling window → generate findings, confidence scores, recommendations
- Reviewer sees Review/HITL page → reads agent findings → approves or rejects

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- After approval → Dashboard shows analytics, anomaly trends, token costs, health score
 - Alerts page shows all triggered alerts from agents and system monitors
 - Audit Trail records every event, agent execution, user action, and system change

2. Tech Stack

Layer	Technology & Version
Framework	React 18 + TypeScript
Build Tool	Vite 6
Styling	Tailwind CSS 3 + CSS Custom Properties (design tokens)
UI Components	Lucide React (icons)
Charts	Recharts 2.12
Fonts	Space Grotesk · JetBrains Mono · DM Sans (Google Fonts)
Routing	React Router v6
State	React useState / useContext
HTTP Client	Native fetch() with async/await

3. Design System & Global UI Conventions

3.1 Color Tokens (CSS Custom Properties)

Token	Hex Value — Usage
--bg-primary	#060D1A — Main page background
--bg-secondary	#0A1628 — Section/panel background
--bg-card	#0D1E35 — Card background
--border-color	#1A3050 — Default border
--border-bright	#1E4080 — Hover/active border
--text-primary	#E8F4FF — Main text
--text-secondary	#8BACC8 — Secondary text
--text-muted	#4A6FA5 — Muted/placeholder text
--accent-blue	#3B82F6 — Primary CTA, active states
--accent-cyan	#06B6D4 — Highlights
--accent-green	#10B981 — Success, approved
--accent-orange	#F59E0B — Warnings
--accent-red	#EF4444 — Errors, critical
--accent-purple	#7C3AED — AI/synthetic elements

3.2 Typography

Font	Usage
Space Grotesk	All headings (h1–h6), labels, titles, buttons
DM Sans	Body text, paragraphs, descriptions
JetBrains Mono	Run IDs, timestamps, code, metrics, badge text

3.3 Layout Structure

All authenticated pages use a fixed two-panel layout:

- Left sidebar: 220px fixed width, dark navy (#0A1628), contains logo + nav groups + system status
- Right content: flex-1, scrollable, contains topbar (56px) + page-specific content

The topbar always contains: page title + subtitle (left), system status badge + datetime + bell icon + user avatar (right).

3.4 Global Components

- Sidebar — logo, MAIN/PIPELINE/OBSERVABILITY/GOVERNANCE nav groups, current run widget at bottom
- Topbar — page heading, System Healthy badge, datetime, notification bell with unread count, user chip with role
- StatusBadge — Approved (green), Pending Review (amber), Rejected (red), with dot indicator
- Glass card — `rgba(13,30,53,0.8)` backdrop-blur border, 12px radius
- Input field — `rgba(6,20,40,0.8)` bg, border-color border, blue focus ring, muted placeholder
- btn-primary — gradient `#2563EB→#3B82F6`, hover lift + glow shadow
- btn-secondary — bg-card, border-color border, hover blue border

4. Page 1 — Login & Signup

4.1 Route

/login — Public, unauthenticated

4.2 Layout

Full-viewport two-column layout (no sidebar):

- Left panel (flex-1): logo top-left, hero headline, shield illustration (animated float), 4 feature cards, privacy note, footer links
- Right panel (480px): glass auth card, switches between Login form and Signup form via tab state

4.3 Login Form

- v4.0 Enhanced badge (green pill, top-left of left panel)
- Heading: "Welcome Back" | Subtext: "Sign in to continue to your AI Operations Command Center"
- Username field — icon: User, placeholder: "Enter your username"
- Password field — icon: Lock, placeholder: "Enter your password", show/hide toggle (Eye/EyeOff)
- "Forgot Password?" link — right-aligned, blue text
- "Sign In" button — btn-primary, full width, LogIn icon, form submit
- OR divider
- "Sign in with Google" button — btn-secondary, Google SVG icon
- "Don't have an account? Sign Up" — text link switches to signup form

4.4 Signup Form

- Heading: "Create Account"
- Full Name field — User icon
- Username field — User icon
- Email field — Mail icon
- Password field — Lock icon, show/hide toggle
- Role dropdown — 7 roles: Clinical Data Manager · Principal Investigator · Data Scientist · Regulatory Affairs Specialist · Clinical Research Associate · System Administrator · Biostatistician
- "Create Account" button — btn-primary, UserPlus icon
- "Already have an account? Sign In" — switches back to login

4.5 Feature Cards (Left Panel, 4-column grid)

Card	Icon / Title / Description
1	 Secure & Compliant — Enterprise-grade security

2	 Real-time Streaming — Kafka-powered pipeline
3	 AI Agents — 5 specialized agents working for you
4	 Human-in-the-Loop — Clinical experts in control

4.6 API for Login & Signup Page

POST /auth/login

POST /auth/login	
Trigger	User clicks "Sign In" button with valid username and password.
Request	<pre>{ "username": "admin", "password": "admin123" }</pre>
Response	<pre>HTTP 200 OK { "success": true, "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "user": { "id": "USR_001", "username": "admin", "fullName": "Dr. Sarah Johnson", "email": "sarah.johnson@clinicaltrials.ai", "role": "Clinical Data Manager", "avatar_initials": "SJ" } }</pre> <pre>HTTP 401 Unauthorized { "success": false, "error": "Invalid credentials" }</pre>
Notes	Token stored in AuthContext / sessionStorage. All subsequent requests send Authorization: Bearer <token>.

POST /auth/signup

POST /auth/signup	
Trigger	User completes signup form and clicks "Create Account".
Request	<pre>{ "fullName": "Dr. Jane Smith", "username": "jane.smith", "email": "jane@clinicaltrials.ai", "password": "SecurePass123!", "role": "Principal Investigator" }</pre>

Response	<pre>HTTP 201 Created { "success": true, "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "user": { "id": "USR_042", "username": "jane.smith", "fullName": "Dr. Jane Smith", "email": "jane@clinicaltrials.ai", "role": "Principal Investigator" } }</pre> <pre>HTTP 409 Conflict { "success": false, "error": "Username already exists" }</pre>
Notes	On success, redirect to /overview. On conflict, show inline error below username field.

POST /auth/logout

POST /auth/logout	
Trigger	User clicks Logout button in topbar.
Request	<pre>Headers: Authorization: Bearer <token> Body: {}</pre>
Response	<pre>HTTP 200 OK { "success": true, "message": "Logged out successfully" }</pre>
Notes	Clear token from AuthContext, redirect to /login.

POST /auth/forgot-password

POST /auth/forgot-password	
Trigger	User clicks "Forgot Password?" link and submits email.
Request	<pre>{ "email": "sarah.johnson@clinicaltrials.ai" }</pre>
Response	<pre>HTTP 200 OK { "success": true, "message": "Reset instructions sent if email exists" }</pre>
Notes	Always return 200 to prevent email enumeration.

5. Page 2 — Overview (Landing)

5.1 Route

/overview — Public landing + authenticated home

5.2 Layout

Sidebar + main content area. Topbar right side shows "Login" link when unauthenticated, or user chip + logout when authenticated.

5.3 Sections

- Hero banner: v4.0 badge, "Agentic AI Data Observability System" headline, gradient subtitle, tagline, shield illustration (animate-float), 5 pill tabs
- "How It Works" — 6 numbered steps with arrows: Input Data → Pre-Ingest Gate → Kafka Streaming → AI Agents → Human Review → Dashboard
- 3-column grid: (1) 5 Observability Pillars cards, (2) System Architecture diagram, (3) What's New in v4.0 checklist
- CTA footer: "Designed for Life Sciences. Built for Trust." + "Get Started →" button

5.4 5 Observability Pillars

Pillar	Icon / Description
Freshness	 Ensure data is up-to-date and within expected time thresholds
Volume	 Monitor data volume and detect unexpected drops or spikes
Schema	 Validate schema consistency and detect breaking changes
Distribution	 Detect data drift and distribution changes over time
Lineage	 Track data lineage and end-to-end pipeline traceability

5.5 API for Overview Page

GET /system/status

```
GET /system/status
```

Trigger	Page load — checks system health for the status widget.
Request	No body. Headers: Authorization: Bearer <token> (optional)
Response	<pre>HTTP 200 OK { "status": "operational", "services": { "kafka": "healthy", "agents": "healthy", "api": "healthy", "storage": "healthy" }, "version": "4.0 Enhanced", "uptime_seconds": 86400 }</pre>
Notes	Shown in sidebar "System Status" widget. Poll every 30 seconds.

GET /pipeline/runs/recent

GET /pipeline/runs/recent?limit=5	
Trigger	Overview page loads recent activity for the "Get Started" CTA section.
Request	No body.
Response	<pre>HTTP 200 OK { "runs": [{ "run_id": "RUN_20250520_001", "status": "Approved", "input_mode": "Upload", "rows": 500, "started_at": "2025-05-20T11:20:00Z", "completed_at": "2025-05-20T11:45:30Z" }], "total": 128 }</pre>

6. Page 3 — Pipeline

6.1 Route

/pipeline — Protected (requires authentication)

6.2 Layout

Sidebar + topbar + two-column content: left column (flex-1) for input/trigger/progress, right column (320px fixed) for validation gate.

6.3 Section 1 — Choose Data Input Mode

Three mode cards. Selecting a mode dynamically replaces the "Mode Detail" section below the cards.

- Mode 1 — User Upload (blue, Upload icon): file drag-drop zone, uploaded file pill, data summary table, schema preview (first 5 columns), Preview Data / Schema Mapper buttons
- Mode 2 — Synthetic Generator (purple, Cpu icon): scenario dropdown (8 named scenarios), row count input (100–10,000), Generate Dataset button, generated dataset summary card
- Mode 3 — External API (green, Globe icon): API URL input, auth type (Bearer/API Key/None), poll interval (5s/30s/60s), token field (shown when auth selected), max records, Test Connection button

6.4 Section 2 — Preflight Validation Gate (right column)

- Summary counters: Hard Blocks (red, XCircle icon) | Soft Warnings (amber, AlertTriangle icon)
- Hard Blocks list: expandable rows with ChevronDown, error message + detail on expand
- "Fix Hard Blocks to Enable Run" CTA — disabled (gray) when blocks exist, green "All Clear" when none
- "Run Checks" button in section header triggers POST /pipeline/preflight/{run_id}

6.5 Section 3 — Trigger Pipeline

- Run Configuration: Run Name, Window Size (events), Inter-event Delay (ms), Description
- "Run Pipeline" button — large blue, Play icon, disabled when hard blocks exist or already running
- "Reset" button — resets stages to pending state
- Active Run panel: shows run ID + spinning icon when running, "No active run / Pipeline is idle" when idle
- Previous Runs table: Run ID (blue mono) | Input Mode | Rows | StatusBadge | Started At | ExternalLink action

6.6 Section 4 — Pipeline Progress

7-stage linear progress indicator with stage icons, status (pending/active/completed/failed), spinner when active.

Stage	Icon
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1. Input Discovery	Database
2. Entry Validation	CheckCircle
3. Preprocessing	Settings
4. PII/PHI Masking	Shield
5. AI Agents	Bot
6. Incident Report	FileText
7. Awaiting Review	User

6.7 API for Pipeline Page

GET /pipeline/runs

GET /pipeline/runs?page=1&limit=10	
Trigger	Pipeline page loads — fetches previous runs table.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "runs": [{ "run_id": "RUN_20250520_001", "input_mode": "Upload", "rows": 500, "status": "Approved", "started_at": "2025-05-20T11:20:00Z", "completed_at": "2025-05-20T11:45:30Z", "description": "User upload run - clinical_trial_data_sample.csv" }], "total": 128, "page": 1, "limit": 10 }</pre>

POST /pipeline/upload

POST /pipeline/upload	
Trigger	User drops a CSV/JSON file onto the upload zone or selects via file browser.
Request	<pre>Content-Type: multipart/form-data { "file": <binary CSV/JSON data>, "run_id": "RUN_20250520_001" (optional - generated server-side if absent)</pre>

	<pre>}</pre>
Response	<pre>HTTP 200 OK { "run_id": "RUN_20250520_005", "filename": "clinical_trial_data_sample.csv", "file_size_mb": 2.45, "total_rows": 500, "total_columns": 10, "detected_format": "CSV", "encoding": "UTF-8", "columns": [{ "name": "patient_id", "type": "string", "sample": "P00123", "null_count": 12, "null_pct": 2.4 }, { "name": "event_date", "type": "datetime", "sample": "2025-05-20 10:15:00", "null_count": 0, "null_pct": 0.0 }, { "name": "age", "type": "integer", "sample": "45", "null_count": 0, "null_pct": 0.0 }, { "name": "gender", "type": "string", "sample": "Male", "null_count": 0, "null_pct": 0.0 }, { "name": "trial_id", "type": "string", "sample": "TRIAL_001", "null_count": 0, "null_pct": 0.0 }], "file_hash": "a3f2bc1d4e5678901234567890abcdef", "saved_to": "data/uploads/RUN_20250520_005_clinical_trial_data_sample.csv" }</pre>
Notes	UI shows schema preview table (first 5 columns shown, "View All 10 Columns" button for rest). File pill shows name, size, row count.

POST /pipeline/generate-synthetic

POST /pipeline/generate-synthetic	
Trigger	User selects scenario + row count and clicks "Generate Dataset".
Request	<pre>{ "scenario": "high_nulls", "rows": 500, "null_rate": null, "outlier_pct": null, "date_drift_days": null, "duplicate_rate": null }</pre>
Response	<pre>HTTP 200 OK { "run_id": "RUN_20250520_006", "scenario": "high_nulls", "rows_generated": 500, "columns": 10, "saved_to": "data/clinical/clinical_trial_data.csv", "generated_at": "2025-05-20T14:30:00Z", "preview": { "null_rate_avg": 0.25, "outlier_pct": 0.0, "severity_distribution": { "critical": 33, "high": 34, "medium": 33, "low": 0 } } }</pre>

Notes	UI shows "Dataset Generated Successfully" card with scenario name, row count, file path, and timestamp.
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POST /pipeline/test-api-connection

POST /pipeline/test-api-connection	
Trigger	User fills API config form and clicks "Test Connection".
Request	<pre>{ "url": "https://api.hospital-data.example.com/trials/events", "auth_type": "Bearer", "token": "eyJhbGciOiJIUzI1NiIs... ", "poll_interval_seconds": 30, "max_records_per_poll": 500 }</pre>
Response	<pre>HTTP 200 OK { "connected": true, "endpoint": "https://api.hospital-data.example.com/trials/events", "auth_type": "Bearer", "response_format": "JSON", "avg_latency_ms": 142, "sample_record_count": 12, "field_names": ["patient_id","event_date","age","gender","trial_id","glucose_level","v ital_signs","medication","severity","visit_date"] }</pre> <pre>HTTP 400 Bad Request { "connected": false, "error": "Connection refused: host unreachable" }</pre>
Notes	UI shows "Connection Successful" card with latency and field names on 200; shows error banner on 400.

POST /pipeline/preflight/{run_id}

POST /pipeline/preflight/{run_id}	
Trigger	User clicks "Run Checks" button OR automatically triggered after file upload.
Request	<pre>{ "run_id": "RUN_20250520_005" }</pre>
Response	<pre>HTTP 200 OK { "run_id": "RUN_20250520_005", "passed": false, "checked_at": "2025-05-20T14:32:00Z", "row_count": 500, "hard_blocks": [{ "id": "hb_001", "check": "missing_required_values", "column": "patient_id", "message": "Missing required values in column: patient_id", "detail": "12 rows (2.4%) have null values" },], }</pre>

	<pre>{ "id": "hb_002", "check": "invalid_date_format", "column": "event_date", "message": "Invalid date format in column: event_date", "detail": "5 rows contain invalid date values" }, "soft_warnings": [{ "id": "sw_001", "check": "high_null_rate", "column": "glucose_level", "message": "High null rate in column: glucose_level", "detail": "Null rate: 8.2% (Recommended < 5%)" }, { "id": "sw_002", "check": "duplicate_ids", "column": "patient_id", "message": "Duplicate patient_id detected", "detail": "8 duplicate IDs found" }, { "id": "sw_003", "check": "stale_data", "column": "visit_date", "message": "Stale data detected", "detail": "12 records older than 180 days" }], "cross_field_violations": [] }</pre>
Notes	Hard blocks disable Run Pipeline button. Soft warnings show but do not block. "Fix Hard Blocks to Enable Run" CTA remains disabled until hard_blocks is empty.

POST /pipeline/run

POST /pipeline/run	
Trigger	User clicks "Run Pipeline" button (only enabled when preflight passed with zero hard blocks).
Request	<pre>Headers: X-API-Key: <pipeline_api_key> { "run_id": "RUN_20250520_005", "run_name": "Clinical Trial Pipeline Run", "input_mode": "upload", "window_size": 500, "inter_event_delay_ms": 50, "description": "User upload run - clinical_trial_data_sample.csv" }</pre>
Response	<pre>HTTP 202 Accepted { "run_id": "RUN_20250520_005", "status": "running", "started_at": "2025-05-20T14:35:00Z", "current_stage": "Input Discovery", "stage_index": 1, "total_stages": 7 }</pre>

	<pre>HTTP 429 Too Many Requests { "error": "Too many requests - wait before retrying", "retry_after_seconds": 60 }</pre>
Notes	Requires X-API-Key header. Rate limited to 3 runs/minute per IP. On 202, frontend polls GET /pipeline/status/{run_id} every 2 seconds to update progress stages.

GET /pipeline/status/{run_id}

GET /pipeline/status/{run_id}	
Trigger	Polled every 2 seconds after Run Pipeline starts, until status is "completed" or "failed".
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "run_id": "RUN_20250520_005", "status": "running", "current_stage": "AI Agents", "stage_index": 5, "total_stages": 7, "stages": [{ "num": 1, "label": "Input Discovery", "status": "completed", "duration_ms": 1200 }, { "num": 2, "label": "Entry Validation", "status": "completed", "duration_ms": 890 }, { "num": 3, "label": "Preprocessing", "status": "completed", "duration_ms": 2300 }, { "num": 4, "label": "PII/PHI Masking", "status": "completed", "duration_ms": 1100 }, { "num": 5, "label": "AI Agents", "status": "active", "duration_ms": null }, { "num": 6, "label": "Incident Report", "status": "pending", "duration_ms": null }, { "num": 7, "label": "Awaiting Review", "status": "pending", "duration_ms": null }], "events_processed": 342, "started_at": "2025-05-20T14:35:00Z" }</pre>
Notes	Stop polling when status = "completed" or "failed". On "completed", show Awaiting Review stage as final state.

POST /pipeline/reset

POST /pipeline/reset	
Trigger	User clicks "Reset" button on pipeline page.
Request	<pre>{ "run_id": "RUN_20250520_005" }</pre>
Response	<pre>HTTP 200 OK { "success": true, "message": "Pipeline state reset successfully" }</pre>
Notes	Resets all stage statuses to "pending" in the UI. Does not delete the run from history.

7. Page 4 — Streaming Pipeline

7.1 Route

/streaming — Protected. Real-time page, polls every 2s for live data.

7.2 Layout

Sidebar + topbar. Topbar shows "LIVE" badge (green pulse) next to page title. Content is a dense grid of monitoring panels.

7.3 Sections

- Pipeline flow diagram: Hospital API Simulator → Kafka Producer → Kafka Topic → Kafka Consumer → Observability Engine (each node shows LIVE badge)
- Pipeline Status sidebar (top-right): RUNNING status, Uptime, Events Processed, Events/Sec avg, Last Event Time
- Kafka Producer Status: Records Sent, Send Rate (msg/s), Last Sent, No Errors indicator
- Kafka Consumer Status: Records Consumed, Consumer Rate (msg/s), Consumer Lag (avg), Last Consumed
- Topic Status: Topic Name (clinical_trials), Partitions, Replication Factor, Under Replicated
- Real-time Event Progress: donut chart (Processed vs Pending), % completed
- Live Consumer Lag chart (line chart, last 5 minutes, orange line, red threshold dashed)
- Events Per Second Throughput chart (line chart, blue, last 5 minutes)
- Stream Window Monitoring: Current Window, Window Start/End, Events in Window, Window Size, Rolling Metrics Status
- AI Agent Processing Status table: Agent | Status | Last Run | Confidence | Findings
- Recent Stream Events table: Event ID | Event Type | Time | Status (Processed/Error)
- Live AI Findings Feed: timestamped agent findings with severity icon
- Streaming Health Overview footer bar: Producer/Topic/Consumer/Lag/Stream health indicators

7.4 API for Streaming Pipeline Page

GET /streaming/status/{run_id}

GET /streaming/status/{run_id}	
Trigger	Page load + polled every 2 seconds while pipeline is running.
Request	No body. Auth required.

Response	<pre>HTTP 200 OK { "run_id": "RUN_20250520_001", "pipeline_status": "RUNNING", "uptime_seconds": 9330, "events_processed": 128542, "events_per_sec_avg": 18.6, "last_event_time": "2025-05-20T12:45:30Z", "producer": { "status": "ONLINE", "records_sent": 128542, "send_rate_msg_per_sec": 18.6, "last_sent": "2025-05-20T12:45:30Z", "errors": 0 }, "consumer": { "status": "ONLINE", "records_consumed": 128120, "consumer_rate_msg_per_sec": 18.5, "consumer_lag_avg": 422, "last_consumed": "2025-05-20T12:45:30Z", "errors": 0 }, "topic": { "name": "clinical_trials", "status": "HEALTHY", "partitions": 12, "replication_factor": 3, "under_replicated": 0 }, "progress": { "total_target_events": 150000, "events_processed": 128542, "events_pending": 21458, "pct_complete": 85.7 } }</pre>
Notes	All live metrics panels update from this single endpoint every 2 seconds.

GET /streaming/lag-history/{run_id}

GET /streaming/lag-history/{run_id}?window=5m	
Trigger	Fetches on load and polled every 5 seconds to update the Consumer Lag line chart.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "run_id": "RUN_20250520_001", "window": "5m", "lag_threshold": 1000, "data_points": [{ "timestamp": "2025-05-20T12:40:30Z", "consumer_lag": 380 }, { "timestamp": "2025-05-20T12:41:00Z", "consumer_lag": 410 }, { "timestamp": "2025-05-20T12:41:30Z", "consumer_lag": 445 }, { "timestamp": "2025-05-20T12:42:00Z", "consumer_lag": 422 }] }</pre>
Notes	Recharts LineChart. X-axis: timestamps. Y-axis: lag value. Orange line for avg lag, red dashed for threshold.

GET /streaming/throughput-history/{run_id}

GET /streaming/throughput-history/{run_id}?window=5m	
Trigger	Polled every 5 seconds to update the Events/Sec throughput line chart.
Request	No body. Auth required.
Response	<pre> HTTP 200 OK { "run_id": "RUN_20250520_001", "window": "5m", "avg_msg_per_sec": 18.6, "data_points": [{ "timestamp": "2025-05-20T12:40:30Z", "events_per_sec": 17.2 }, { "timestamp": "2025-05-20T12:41:00Z", "events_per_sec": 19.1 }, { "timestamp": "2025-05-20T12:42:00Z", "events_per_sec": 18.6 }] }</pre>
Notes	Blue line chart. Avg value shown as badge top-right of chart.

GET /streaming/events/recent/{run_id}

GET /streaming/events/recent/{run_id}?limit=5	
Trigger	Polled every 3 seconds to update the Recent Stream Events table.
Request	No body. Auth required.
Response	<pre> HTTP 200 OK { "events": [{ "event_id": "EVT_128542", "event_type": "Lab_Results", "time": "2025-05-20T12:45:30Z", "status": "Processed" }, { "event_id": "EVT_128541", "event_type": "Patient_Vitals", "time": "2025-05-20T12:45:29Z", "status": "Processed" }, { "event_id": "EVT_128540", "event_type": "Medication", "time": "2025-05-20T12:45:29Z", "status": "Processed" }] }</pre>

GET /streaming/agents/status/{run_id}

GET /streaming/agents/status/{run_id}	
Trigger	Polled every 3 seconds to update the AI Agent Processing Status table.
Request	No body. Auth required.
Response	<pre> HTTP 200 OK { "agents": [{ "name": "Data Quality Agent", "status": "COMPLETED", "last_run": "2025-05-20T12:45:22Z", "confidence": 92, "findings": "3 anomalies detected" },] }</pre>

	<pre>{ "name": "Log Analysis Agent", "status": "COMPLETED", "last_run": "2025-05-20T12:45:24Z", "confidence": 88, "findings": "2 patterns identified" }, { "name": "RCA Agent", "status": "RUNNING", "last_run": null, "confidence": null, "findings": "Investigating root cause..." }, { "name": "Recommendation Agent", "status": "PENDING", "last_run": null, "confidence": null, "findings": "Waiting for RCA" }, { "name": "Compliance Agent", "status": "PENDING", "last_run": null, "confidence": null, "findings": "Waiting for recommendations" } }</pre>
--	--

GET /streaming/ai-findings/live/{run_id}

GET /streaming/ai-findings/live/{run_id}?limit=5	
Trigger	Polled every 5 seconds to update the Live AI Findings Feed.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "findings": [{ "id": "F001", "severity": "critical", "message": "High null rate detected in glucose_level", "agent": "Data Quality Agent", "timestamp": "2025-05-20T12:45:22Z" }, { "id": "F002", "severity": "warning", "message": "Unusual spike in glucose_level values", "agent": "Data Quality Agent", "timestamp": "2025-05-20T12:45:22Z" }, { "id": "F003", "severity": "info", "message": "Potential ingestion delay from API source", "agent": "RCA Agent", "timestamp": "2025-05-20T12:45:28Z" }] }</pre>
Notes	Severity icons: critical=red XCircle, warning=amber AlertTriangle, info=blue Info.

GET /streaming/window/status/{run_id}

GET /streaming/window/status/{run_id}	
Trigger	Polled every 2 seconds to update Stream Window Monitoring panel.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "current_window": "Last 1 Minute", "window_start": "2025-05-20T12:44:30Z", "window_end": "2025-05-20T12:45:30Z", "events_in_window": 1116, "window_size": "1 Min", "rolling_metrics_status": "ACTIVE" }</pre>

8. Page 5 — Review / HITL

8.1 Route

/review — Protected. Only accessible to users with Clinical Data Manager role (or equivalent).

8.2 Layout

Sidebar + topbar. Topbar shows "Review Pending" badge (amber), Run ID, datetime, user chip. Two-column: main content (left, ~70%) + review sidebar (right, ~30%).

8.3 Sections

- Run summary bar: Input Mode card, Total Records card, Pipeline Status card (Agents Completed), Started At + Completed At card, Health Score card (82/100 Good)
- AI Agent Findings tabs: Data Quality Agent | Log Analysis Agent | RCA Agent | Recommendation Agent | Compliance Agent — each tab shows Summary, Key Findings table, Evidence block, Recommendation list, Insight panel, Confidence Score badge
- All Agent Confidence Overview: 5 mini-cards with confidence score gauge, sparkline, agent name, score/100, qualitative label
- Attention Required banner — shown when any agent confidence < 60 (amber, AlertTriangle)
- Run Artifacts: Incident Report (JSON, 128KB) | Observability Metrics (JSON, 256KB) | ETL Log (CSV, 89KB) | Agent Responses (JSON, 312KB) | Masked Output (JSON, 145KB)

8.4 Review Sidebar

- Review Summary stats: Total Agents, Completed, High Confidence (≥60), Low Confidence (<60), Critical Issues
- Overall Confidence donut gauge: score/100, label (Good/Caution/Poor)
- Token & Cost Summary table: per-agent Tokens + Cost (USD), total row
- Human Review Actions: Reviewer ID (pre-filled from auth), Review Notes textarea
- "✓ Approve Report" — large green button, POST /review/approve
- "✗ Reject & Request Revision" — red button, POST /review/reject
- "Escalate to Compliance Officer" — checkbox

8.5 Key Findings Table (per agent tab)

Column	Data
Issue	Finding name (e.g. "Null rate - glucose_level")
Severity	Critical/High/Medium/Low badge
Affected	"158 / 500" (affected rows / total)

Percentage	"31.6%"
Trend	Arrow icon (up=worsening, sideways=stable)

8.6 API for Review / HITL Page

GET /review/pending

GET /review/pending	
Trigger	Review page loads — fetches runs awaiting review.
Request	No body. Auth required (CDM role).
Response	<pre>HTTP 200 OK { "pending_runs": [{ "run_id": "RUN_20250520_001", "input_mode": "User Upload (CSV)", "filename": "clinical_trial_data_sample.csv", "total_records": 500, "window_size": 500, "pipeline_status": "Agents Completed", "started_at": "2025-05-20T12:43:00Z", "completed_at": "2025-05-20T12:45:00Z", "health_score": 82, "health_label": "Good" }] }</pre>

GET /review/findings/{run_id}

GET /review/findings/{run_id}	
Trigger	Review page loads for a specific run — fetches all agent findings.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "run_id": "RUN_20250520_001", "agents": [{ "name": "Data Quality Agent", "status": "Completed", "confidence": 92, "confidence_label": "High", "flag_for_review": false, "summary": "The dataset contains multiple data quality issues that may impact downstream analysis.", "key_findings": [</pre>

	<pre> { "issue": "Null rate - glucose_level", "severity": "Critical", "affected": 158, "total": 500, "percentage": 31.6, "trend": "up" }, { "issue": "Outliers - glucose_level", "severity": "High", "affected": 42, "total": 500, "percentage": 8.4, "trend": "up" }, { "issue": "Invalid event_date", "severity": "Medium", "affected": 12, "total": 500, "percentage": 2.4, "trend": "stable" }, { "issue": "Duplicate patient_id", "severity": "Medium", "affected": 8, "total": 500, "percentage": 1.6, "trend": "stable" }, { "issue": "Stale data (>180 days)", "severity": "Low", "affected": 20, "total": 500, "percentage": 4.0, "trend": "up" }], "evidence": "Column: glucose_level Null Count: 156 Null Percentage: 31.6% Example Null Rows: 12, 45, 67, 98, 123...", "recommendations": ["Validate source system for missing glucose values", "Implement not-null rule for glucose_level", "Review data collection process", "Monitor null rate trend"], "insight": "High null rate in glucose values can impact analysis accuracy." }], "overall_confidence": 76, "attention_required": true, "attention_message": "RCA Agent confidence is below 60%. Manual review strongly recommended.", "review_summary": { "total_agents": 5, "completed": 5, "high_confidence": 4, "low_confidence": 1, "critical_issues": 2 } }</pre>
Notes	Agent findings are static (already computed). Render each agent as a tab. Show Attention Required banner if attention_required = true.

GET /review/token-usage/{run_id}

GET /review/token-usage/{run_id}	
Trigger	Review page loads — fetches token usage for cost summary sidebar.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "run_id": "RUN_20250520_001", "agents": [{ "name": "Data Quality Agent", "input_tokens": 1842, "output_tokens": 308, "total_tokens": 2150, "cost_usd": 0.028 }, { "name": "Log Analysis Agent", "input_tokens": 1105, "output_tokens": 215, "total_tokens": 1320, "cost_usd": 0.017 }, { "name": "RCA Agent", "input_tokens": 2340, "output_tokens": 520, "total_tokens": 2860, "cost_usd": 0.037 }, { "name": "Recommendation Agent", "input_tokens": 980, "output_tokens": 200, "total_tokens": 1180, "cost_usd": 0.016 }, { "name": "Compliance Agent", "input_tokens": 1280, "output_tokens": 260, "total_tokens": 1540, "cost_usd": 0.020 }] }</pre>

	<pre>], "total_tokens": 9050, "total_cost_usd": 0.117 }</pre>
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POST /review/approve

POST /review/approve	
Trigger	Reviewer clicks "✓ Approve Report" button.
Request	<pre>Headers: X-API-Key: <pipeline_api_key> { "run_id": "RUN_20250520_001", "reviewer_id": "CDM_01", "notes": "Reviewed all agent outputs. Critical issues acknowledged. Recommendations are appropriate.", "escalate_to_compliance": false }</pre>
Response	<pre>HTTP 200 OK { "success": true, "run_id": "RUN_20250520_001", "status": "Approved", "approved_by": "CDM_01", "approved_at": "2025-05-20T12:50:00Z", "dashboard_url": "/dashboard/RUN_20250520_001" }</pre>
Notes	On success, redirect to /dashboard/RUN_20250520_001. Audit trail event is written automatically by backend.

POST /review/reject

POST /review/reject	
Trigger	Reviewer clicks "✗ Reject & Request Revision" button.
Request	<pre>Headers: X-API-Key: <pipeline_api_key> { "run_id": "RUN_20250520_001", "reviewer_id": "CDM_01", "reason": "Data quality issues in glucose_level require source system investigation before approval.", "escalate_to_compliance": true }</pre>
Response	<pre>HTTP 200 OK { "success": true, "run_id": "RUN_20250520_001", "status": "Rejected", "rejected_by": "CDM_01", "rejected_at": "2025-05-20T12:52:00Z" }</pre>
Notes	Run status set to Rejected in sidebar. Option to re-run pipeline shown. Audit trail event written.

GET /review/artifacts/{run_id}

GET /review/artifacts/{run_id}	
Trigger	Review page loads — fetches downloadable run artifacts list.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "run_id": "RUN_20250520_001", "artifacts": [{ "name": "Incident Report", "type": "JSON", "size_kb": 128, "path": "output/runs/RUN_20250520_001/incident_report.json" }, { "name": "Observability Metrics", "type": "JSON", "size_kb": 256, "path": "output/runs/RUN_20250520_001/metrics.json" }, { "name": "ETL Log", "type": "CSV", "size_kb": 89, "path": "output/runs/RUN_20250520_001/etl_log.csv" }, { "name": "Agent Responses", "type": "JSON", "size_kb": 312, "path": "output/runs/RUN_20250520_001/agent_responses.json" }, { "name": "Masked Output", "type": "JSON", "size_kb": 145, "path": "output/runs/RUN_20250520_001/masked_output.json" }] }</pre>

9. Page 6 — Dashboard

9.1 Route

/dashboard or /dashboard/{run_id} — Protected. Data only shown for Approved runs.

9.2 Layout

Sidebar + topbar with hamburger menu toggle. Topbar right side: System Healthy badge, date picker, "Last 24 Hours" dropdown, notification bell, user chip.

9.3 KPI Cards (top row, 5 cards)

Card	Metric / Color / Change indicator
Total Runs	128 ▲ 12.6% vs yesterday (blue icon)
Completed Runs	116 ▲ 15.3% vs yesterday (green checkmark)
Anomalies Detected	48 ▼ 8.2% vs yesterday (amber warning)
Critical Issues	5 ▼ 16.7% vs yesterday (red shield)
Avg. Confidence Score	82/100 ▲ 4.3% vs yesterday (purple shield)

9.4 Charts & Panels (6 sections in 3×2 grid)

- Pipeline Runs Over Time — Recharts BarChart, stacked Completed (green) / Failed (red) / In Progress (blue), Hourly dropdown
- Anomalies by Severity — Recharts PieChart donut, 4 segments with legend, total in center
- Recent Alerts — list of 5 most recent alerts with severity icon, title, agent, time
- AI Agents Performance — table: Agent | Status | Runs | Avg Confidence | Issues
- Token Usage & Cost (Today) — KPI cluster: Total Tokens, Total Cost, Runs, Cost/Run + donut pie by agent
- Pipeline Health — radial gauge (95% Excellent) + 4 KPIs: Availability, Success Rate, Avg Processing Time, Events Processed
- Anomalies Trend — Recharts LineChart, 4 lines (Critical/High/Medium/Low) by day
- Top Anomaly Types — table: Anomaly Type | Count | %
- Run Status Distribution — Recharts PieChart donut, Completed/Failed/In Progress + KPI stats below

9.5 API for Dashboard Page

GET /dashboard/summary

GET /dashboard/summary?period=24h	
Trigger	Dashboard page loads — fetches all KPI card data.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "period": "24h", "total_runs": 128, "total_runs_change_pct": 12.6, "completed_runs": 116, "completed_runs_change_pct": 15.3, "anomalies_detected": 48, "anomalies_change_pct": -8.2, "critical_issues": 5, "critical_issues_change_pct": -16.7, "avg_confidence_score": 82, "avg_confidence_change_pct": 4.3 }</pre>
Notes	Only counts approved runs.

GET /dashboard/pipeline-runs-over-time

GET /dashboard/pipeline-runs-over-time?period=24h&granularity=hourly	
Trigger	Dashboard page loads + granularity dropdown change.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "granularity": "hourly", "data": [{ "time": "12 AM", "completed": 3, "failed": 0, "in_progress": 1 }, { "time": "4 AM", "completed": 8, "failed": 1, "in_progress": 2 }, { "time": "8 AM", "completed": 22, "failed": 2, "in_progress": 3 }, { "time": "12 PM", "completed": 35, "failed": 1, "in_progress": 4 }, { "time": "4 PM", "completed": 28, "failed": 2, "in_progress": 3 }, { "time": "8 PM", "completed": 20, "failed": 1, "in_progress": 2 }] }</pre>
Notes	Renders as stacked BarChart. Granularity: hourly, daily, weekly.

GET /dashboard/anomalies-by-severity

GET /dashboard/anomalies-by-severity?period=24h	
Trigger	Dashboard page loads.
Request	No body. Auth required.

Response	<pre>HTTP 200 OK { "total": 48, "breakdown": [{ "severity": "Critical", "count": 5, "pct": 10.4, "color": "#EF4444" }, { "severity": "High", "count": 12, "pct": 25.0, "color": "#F59E0B" }, { "severity": "Medium", "count": 18, "pct": 37.5, "color": "#F59E0B" }, { "severity": "Low", "count": 13, "pct": 27.1, "color": "#3B82F6" }], "change_pct": -8.3 }</pre>
Notes	Donut PieChart with total in center. Legend shows count + percentage per severity.

GET /dashboard/agents-performance

GET /dashboard/agents-performance?period=24h	
Trigger	Dashboard page loads.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "agents": [{ "name": "Data Quality Agent", "status": "Completed", "runs": 128, "avg_confidence": 92, "issues": 8 }, { "name": "Log Analysis Agent", "status": "Completed", "runs": 128, "avg_confidence": 88, "issues": 6 }, { "name": "RCA Agent", "status": "Completed", "runs": 128, "avg_confidence": 78, "issues": 12 }, { "name": "Recommendation Agent", "status": "Completed", "runs": 128, "avg_confidence": 85, "issues": 4 }, { "name": "Compliance Agent", "status": "Completed", "runs": 128, "avg_confidence": 89, "issues": 3 }] }</pre>

GET /dashboard/token-usage

GET /dashboard/token-usage?period=24h	
Trigger	Dashboard page loads.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "total_tokens": 1420000, "total_cost_usd": 3.28, "runs": 128, "cost_per_run": 0.026, "by_agent": [{ "name": "Data Quality Agent", "tokens": 420000, "pct": 29.6 }, { "name": "Log Analysis Agent", "tokens": 310000, "pct": 21.8 }, { "name": "RCA Agent", "tokens": 380000, "pct": 26.8 }, { "name": "Recommendation Agent", "tokens": 180000, "pct": 12.7 },] }</pre>

	<pre>{ "name": "Compliance Agent", "tokens": 130000, "pct": 9.1 }]</pre>
Notes	Renders as donut pie + KPI cluster.

GET /dashboard/pipeline-health

GET /dashboard/pipeline-health	
Trigger	Dashboard page loads.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "health_score": 95, "health_label": "Excellent", "availability_pct": 100.0, "success_rate_pct": 95.3, "avg_processing_time_seconds": 138, "events_processed": 128542 }</pre>
Notes	Radial gauge renders health_score. Color: ≥90=green, 70–89=amber, <70=red.

GET /dashboard/anomalies-trend

GET /dashboard/anomalies-trend?period=7d&granularity=daily	
Trigger	Dashboard page loads + granularity change.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "data": [{ "date": "May 14", "critical": 8, "high": 20, "medium": 35, "low": 15 }, { "date": "May 15", "critical": 10, "high": 25, "medium": 40, "low": 18 }, { "date": "May 20", "critical": 5, "high": 12, "medium": 18, "low": 13 }] }</pre>
Notes	4-line chart. Critical=red, High=orange, Medium=amber, Low=blue.

GET /dashboard/top-anomaly-types

GET /dashboard/top-anomaly-types?period=24h&limit=5	
Trigger	Dashboard page loads.
Request	No body. Auth required.

Response	<pre>HTTP 200 OK { "anomaly_types": [{ "type": "High null rate in glucose_level", "count": 18, "pct": 37.5 }, { "type": "Outliers in vital_signs", "count": 12, "pct": 25.0 }, { "type": "Invalid date format in event_date", "count": 8, "pct": 16.7 }, { "type": "Duplicate patient_id", "count": 6, "pct": 12.5 }, { "type": "Stale data (>180 days)", "count": 4, "pct": 8.3 }] }</pre>
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GET /dashboard/run-status-distribution

GET /dashboard/run-status-distribution?period=24h	
Trigger	Dashboard page loads.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "total": 128, "completed": { "count": 116, "pct": 90.6 }, "failed": { "count": 7, "pct": 5.5 }, "in_progress": { "count": 5, "pct": 3.9 }, "change_pct": 12.6 }</pre>
Notes	Donut PieChart. Total in center. Completed=green, Failed=red, In Progress=blue.

GET /dashboard/recent-alerts

GET /dashboard/recent-alerts?limit=5	
Trigger	Dashboard page loads.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "alerts": [{ "id": "ALT_001", "severity": "critical", "title": "High null rate in glucose_level", "agent": "Data Quality Agent", "time": "2025-05-20T12:45:00Z" }, { "id": "ALT_002", "severity": "warning", "title": "Outliers detected in vital_signs", "agent": "Log Analysis Agent", "time": "2025-05-20T12:38:00Z" }] }</pre>

10. Page 7 — Alerts

10.1 Route

/alerts — Protected.

10.2 Layout

Sidebar + topbar. Three-panel: severity summary cards (top), alert table (left ~65%), alert detail panel (right ~35%, slides in on row click).

10.3 Sections

- 4 severity cards: Critical (2) — "Requires immediate attention" + View Critical link | High (5) | Medium (12) | Low (8)
- Total Alerts KPI: 27, Last 24 hours, ▼12.9% vs yesterday
- Filter bar: Search input | All Severities dropdown | All Sources dropdown | All Statuses dropdown | Date range picker
- Alert table: Severity badge | Alert Title + description | Source | Run ID | Time | Status (New/In Progress/Acknowledged/Resolved)
- Pagination: showing 1 to 8 of 27 alerts

10.4 Alert Detail Panel (right, slides in on row click)

- Header: severity badge, "New" badge, Alert Title
- Agent badge (e.g. "Data Quality Agent"), Triggered At, Run ID (clickable link)
- Description: full text
- Impact: text description
- Metrics: Null Rate (31.6%) | Threshold (20%) | Affected Records | Column name
- Recommended Action: text
- Actions: "Acknowledge" button (blue) | "Create RCA" button (outline) | "Escalate" button (red, full width)
- Alert History: count badge + expand

10.5 API for Alerts Page

GET /alerts

GET /alerts?severity=all&source=all&status=all&from=2025-05-19&to=2025-05-20&page=1&limit=10	
Trigger	Page load + filter change + pagination.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "total": 27, "page": 1, "limit": 10, "summary": { "critical": 2, "high": 5, "medium": 12, "low": 8, "total_24h": 27, "change_pct": -12.9 }, "alerts": [{ "id": "ALT_001", "severity": "critical", "title": "High null rate detected in glucose_level", "description": "Null rate: 31.6% (Threshold: 20%)", "source": "Data Quality Agent", "run_id": "RUN_20250520_001", "time": "2025-05-20T12:45:22Z", "status": "New" }, { "id": "ALT_002", "severity": "critical", "title": "Outliers detected in vital_signs", "description": "Outliers: 42 records (Threshold: 30)", "source": "Log Analysis Agent", "run_id": "RUN_20250520_001", "time": "2025-05-20T12:38:15Z", "status": "New" }] }</pre>
Notes	Polled every 10 seconds (new alerts may arrive). Bell icon in topbar shows unread count from this endpoint.

GET /alerts/{alert_id}

GET /alerts/{alert_id}	
Trigger	User clicks a row in the alert table — detail panel slides in.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "id": "ALT_001", "severity": "critical", "is_new": true, "title": "High null rate detected in glucose_level",</pre>

	<pre>"source": "Data Quality Agent", "triggered_at": "2025-05-20T12:45:22Z", "run_id": "RUN_20250520_001", "description": "The column glucose_level has a null rate of 31.6%, which is above the threshold of 20%", "impact": "High impact on data quality and downstream analysis accuracy.", "metrics": { "null_rate_pct": 31.6, "threshold_pct": 20.0, "affected_records": 158, "total_records": 500, "column": "glucose_level" }, "recommended_action": "Review data collection process and validate source system for missing glucose_level values.", "history": [{ "action": "Alert triggered", "by": "system", "at": "2025-05-20T12:45:22Z" }] }</pre>
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POST /alerts/{alert_id}/acknowledge

POST /alerts/{alert_id}/acknowledge	
Trigger	Reviewer clicks "Acknowledge" button in detail panel.
Request	<pre>{ "acknowledged_by": "CDM_01", "note": "Investigating glucose data source" }</pre>
Response	<pre>HTTP 200 OK { "success": true, "alert_id": "ALT_001", "status": "Acknowledged", "acknowledged_by": "CDM_01", "acknowledged_at": "2025-05-20T13:00:00Z" }</pre>
Notes	Status badge in table row updates to "Acknowledged" (teal).

POST /alerts/{alert_id}/escalate

POST /alerts/{alert_id}/escalate	
Trigger	Reviewer clicks "Escalate" button in detail panel.
Request	<pre>{ "escalated_by": "CDM_01", "reason": "Data integrity risk requires compliance officer review" }</pre>
Response	<pre>HTTP 200 OK { "success": true, "alert_id": "ALT_001", "status": "Escalated", }</pre>

	<pre>"escalated_at": "2025-05-20T13:05:00Z" }</pre>
--	---

POST /alerts/{alert_id}/read

POST /alerts/{alert_id}/read	
Trigger	Alert detail panel opens — marks alert as read automatically.
Request	<pre>{ "alert_id": "ALT_001" }</pre>
Response	<pre>HTTP 200 OK { "success": true, "alert_id": "ALT_001", "read": true }</pre>
Notes	Reduces unread count badge in topbar bell icon. Polled by useAlerts hook every 10 seconds.

11. Page 8 — Audit Trail

11.1 Route

/audit — Protected.

11.2 Layout

Sidebar + topbar. Topbar right: date range picker (May 19–May 20), datetime, bell (badge 8), user chip. Two-column: events table (left) + event detail panel (right, slides in on row click).

11.3 KPI Cards (top row, 5 cards)

Card	Metric / Change
Total Events	2,842 ▲ 12.7%
AI Agent Executions	512 ▲ 18.3%
Data Access Events	1,024 ▲ 8.9%
User Actions	752 ▲ 15.2%
Errors	37 ▼ 23.1%

11.4 Filters

- Search input: "Search events, agents, users..."
- All Event Types dropdown: AI Agent Execution | Data Access | User Action | System Event | Alert Triggered | Data Validation | Pipeline Start | User Login
- All Agents dropdown: Data Quality Agent | Log Analysis Agent | RCA Agent | Recommendation Agent | Compliance Agent | all
- All Users dropdown: CDM_01 | system | all
- All Status dropdown: Success | Error | Triggered | all
- Date range picker: start datetime → end datetime

11.5 Events Table

Column	Description
Time	Date + time of event (e.g. "May 20, 2025 12:45:22 PM")
Event Type	Badge: AI Agent Execution (purple) Data Access (blue) User Action (teal) System Event (gray) Alert Triggered (red) Data Validation (green) Pipeline Start (blue) User Login (teal)

Agent / Source	Hyperlink if agent (e.g. "Data Quality Agent (gpt-4o)"), plain text if system
Event Description	First line: short description. Second line: detail (e.g. "Null rate analysis on glucose_level")
User / System	system or CDM_01 etc.
Status	Success (green) Error (red) Triggered (amber)
Run ID	Clickable blue mono link
Actions	Eye icon to open detail panel

11.6 Event Detail Panel

- Header: Event Type badge | Status badge
- Event ID, Time, Run ID (clickable)
- Agent Information: Agent Name, Model Used, Agent Version
- Token Usage: Input Tokens, Output Tokens, Total Tokens, Estimated Cost (USD)
- Event Details: Description, Task, Status, Duration
- Request & Response: "View full prompt" link | "View response" link
- Metadata: Source IP, Environment, Platform, Session ID, User / System

11.7 API for Audit Trail Page

GET /audit/summary

GET /audit/summary?from=2025-05-19&to=2025-05-20	
Trigger	Audit page loads — fetches KPI card data.
Request	No body. Auth required.
Response	HTTP 200 OK { "period": { "from": "2025-05-19T00:00:00Z", "to": "2025-05-20T23:59:59Z" }, "total_events": 2842, "total_events_change_pct": 12.7, "ai_agent_executions": 512, "ai_executions_change_pct": 18.3, "data_access_events": 1024, "data_access_change_pct": 8.9, "user_actions": 752, "user_actions_change_pct": 15.2, "errors": 37, "errors_change_pct": -23.1 }

GET /audit/events

GET /audit/events?event_type=all&agent=all&user=all&status=all&from=2025-05-19T00:00:00Z&to=2025-05-20T23:59:59Z&page=1&limit=10&search=	
Trigger	Page load + filter change + pagination + search.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "total": 2842, "page": 1, "limit": 10, "events": [{ "id": "EVT_20250520_000512", "time": "2025-05-20T12:45:22Z", "event_type": "AI Agent Execution", "event_type_color": "purple", "agent_source": "Data Quality Agent (gpt-4o)", "description": "Completed execution successfully", "detail": "Null rate analysis on glucose_level", "user": "system", "status": "Success", "run_id": "RUN_20250520_001" }, { "id": "EVT_20250520_000511", "time": "2025-05-20T12:45:15Z", "event_type": "Data Access", "event_type_color": "blue", "agent_source": "clinical_trial_data_sample.csv", "description": "File read: clinical_trial_data_sample.csv", "detail": "Rows: 500, Columns: 10", "user": "system", "status": "Success", "run_id": "RUN_20250520_001" }] }</pre>
Notes	Table shows 10 rows per page. Pagination shows "Showing 1 to 10 of 2,842 events".

GET /audit/events/{event_id}

GET /audit/events/{event_id}	
Trigger	User clicks Eye icon on a table row — detail panel slides in.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "id": "EVT_20250520_000512", "event_type": "AI Agent Execution", "status": "Success", "time": "2025-05-20T12:45:22Z", "run_id": "RUN_20250520_001", "agent": { "name": "Data Quality Agent",</pre>

	<pre>"model": "gpt-4o", "version": "1.2.3" }, "token_usage": { "input_tokens": 2153, "output_tokens": 1024, "total_tokens": 3177, "estimated_cost_usd": 0.0072 }, "event_details": { "description": "Completed execution successfully", "task": "Null rate analysis on glucose_level", "status": "Success", "duration_seconds": 18.45 }, "request_response": { "prompt_path": "output/runs/RUN_20250520_001/prompts/data_quality_agent.txt", "response_path": "output/runs/RUN_20250520_001/responses/data_quality_agent.json" }, "metadata": { "source_ip": "127.0.0.1", "environment": "local", "platform": "Linux", "session_id": "sess_8f7a2d3e4b1c", "user": "system" } }</pre>
Notes	Panel shows full detail. "View full prompt" and "View response" are GET links to the file paths.

GET /audit/events/{event_id}/prompt

GET /audit/events/{event_id}/prompt	
Trigger	User clicks "View full prompt" link in detail panel.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "event_id": "EVT_20250520_000512", "prompt": "You are a Data Quality Agent for clinical trial data... [full prompt text here]" }</pre>
Notes	Rendered in a modal or new tab with monospace font.

GET /audit/export

GET /audit/export?from=2025-05-19&to=2025-05-20&format=csv	
Trigger	User clicks an export button (if implemented). Downloads full filtered audit log.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK Content-Type: text/csv</pre>

	<code>Content-Disposition: attachment; filename=audit_trail_2025-05-19_2025-05-20.csv [CSV data stream]</code>
Notes	format can be csv or json. File download triggered in browser.

12. Global Shared Components

12.1 Sidebar Navigation

MAIN section: Overview, Pipeline, Streaming Pipeline, Kafka Status, Event Flow, Metrics Engine
OBSERVABILITY section: 5 Pillars, Anomalies, AI Agents, RCA Insights
GOVERNANCE section: Review (HITL), Audit Trail, Alerts, Compare Runs, PII / PHI Monitor
Bottom: Current Run widget (Run ID, Completed/Running/Pending Review badge, Started/Completed times, View Run Details link)

12.2 Current Run Widget

Shows in sidebar bottom. Polls GET /pipeline/runs/active every 5 seconds.

GET /pipeline/runs/active	
Trigger	Polled every 5 seconds by sidebar Current Run widget.
Request	No body. Auth required.
Response	<pre>HTTP 200 OK { "run_id": "RUN_20250520_001", "status": "Completed", "review_status": "Pending Review", "started_at": "2025-05-20T12:43:12Z", "completed_at": "2025-05-20T12:45:30Z" } HTTP 200 OK (no active run) { "run_id": null, "status": "idle" }</pre>

12.3 Notification Bell

Polls GET /alerts?status=new&limit=99 every 10 seconds. Shows count badge (red circle) on bell icon.

12.4 Authentication Guard

All protected routes check for valid token in AuthContext. If missing or expired, redirect to /login. Pipeline items in sidebar show login modal if user not authenticated.

13. Complete API Reference Index

Method	Endpoint
POST	/auth/login
POST	/auth/signup
POST	/auth/logout
POST	/auth/forgot-password
GET	/system/status
GET	/pipeline/runs/recent
GET	/pipeline/runs
POST	/pipeline/upload
POST	/pipeline/generate-synthetic
POST	/pipeline/test-api-connection
POST	/pipeline/preflight/{run_id}
POST	/pipeline/run
GET	/pipeline/status/{run_id}
POST	/pipeline/reset
GET	/streaming/status/{run_id}
GET	/streaming/lag-history/{run_id}
GET	/streaming/throughput-history/{run_id}
GET	/streaming/events/recent/{run_id}
GET	/streaming/agents/status/{run_id}
GET	/streaming/ai-findings/live/{run_id}
GET	/streaming/window/status/{run_id}
GET	/review/pending
GET	/review/findings/{run_id}
GET	/review/token-usage/{run_id}
POST	/review/approve
POST	/review/reject
GET	/review/artifacts/{run_id}
GET	/dashboard/summary
GET	/dashboard/pipeline-runs-over-time

GET	/dashboard/anomalies-by-severity
GET	/dashboard/agents-performance
GET	/dashboard/token-usage
GET	/dashboard/pipeline-health
GET	/dashboard/anomalies-trend
GET	/dashboard/top-anomaly-types
GET	/dashboard/run-status-distribution
GET	/dashboard/recent-alerts
GET	/alerts
GET	/alerts/{alert_id}
POST	/alerts/{alert_id}/acknowledge
POST	/alerts/{alert_id}/escalate
POST	/alerts/{alert_id}/read
GET	/audit/summary
GET	/audit/events
GET	/audit/events/{event_id}
GET	/audit/events/{event_id}/prompt
GET	/audit/export
GET	/pipeline/runs/active

Auth Legend

Yes = Authorization: Bearer <token> · API-Key = X-API-Key header also required · No = Public endpoint